

# Data transformation and descriptive statistics

CVEN 5999 - Summer 2025

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# Data Organisation in Spreadsheets

# Think, Pair, Share

## Questions

1. Why should you not leave a blank cell in a spreadsheet used for data collection?
2. Which of the 12 rules for data organization was the least comprehensible to you?

- **Think** for 2 minutes
- **Pair** with in break-out rooms for 4 minutes
- **Share** your answer with the class

02:00

 [cven5999-ss25.github.io/website/](https://cven5999-ss25.github.io/website/)

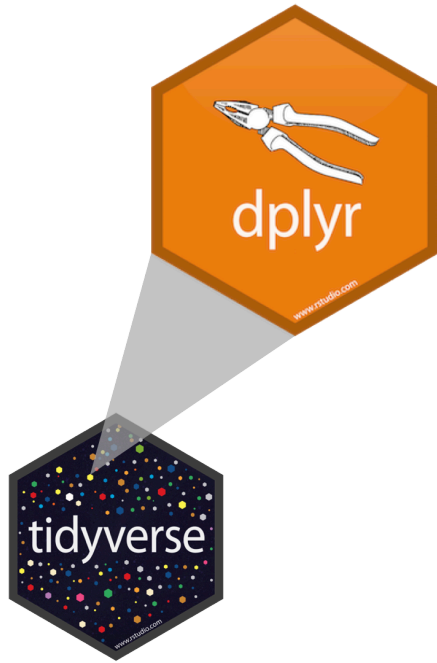
# Learning Objectives (for this week)

1. Learners can apply ten functions from the dplyr R Package to generate a subset of data for use in a table or plot.

# Data wrangling with dplyr

# A grammar of data wrangling...

... based on the concepts of functions as verbs that manipulate data frames



- `select`: pick columns by name
- `arrange`: reorder rows
- `slice`: chooses rows based on location
- `filter`: pick rows matching criteria
- `relocate`: changes the order of the columns
- `mutate`: add new variables
- `summarise`: reduce variables to values
- `group_by`: for grouped operations
- ... (many more)

# dplyr rules

Rules of `dplyr` functions:

- First argument is always a data frame
- Subsequent arguments say what to do with that data frame
- Always return a data frame
- Don't modify in place

# Live Coding Exercise: SDG

## 6.2.1

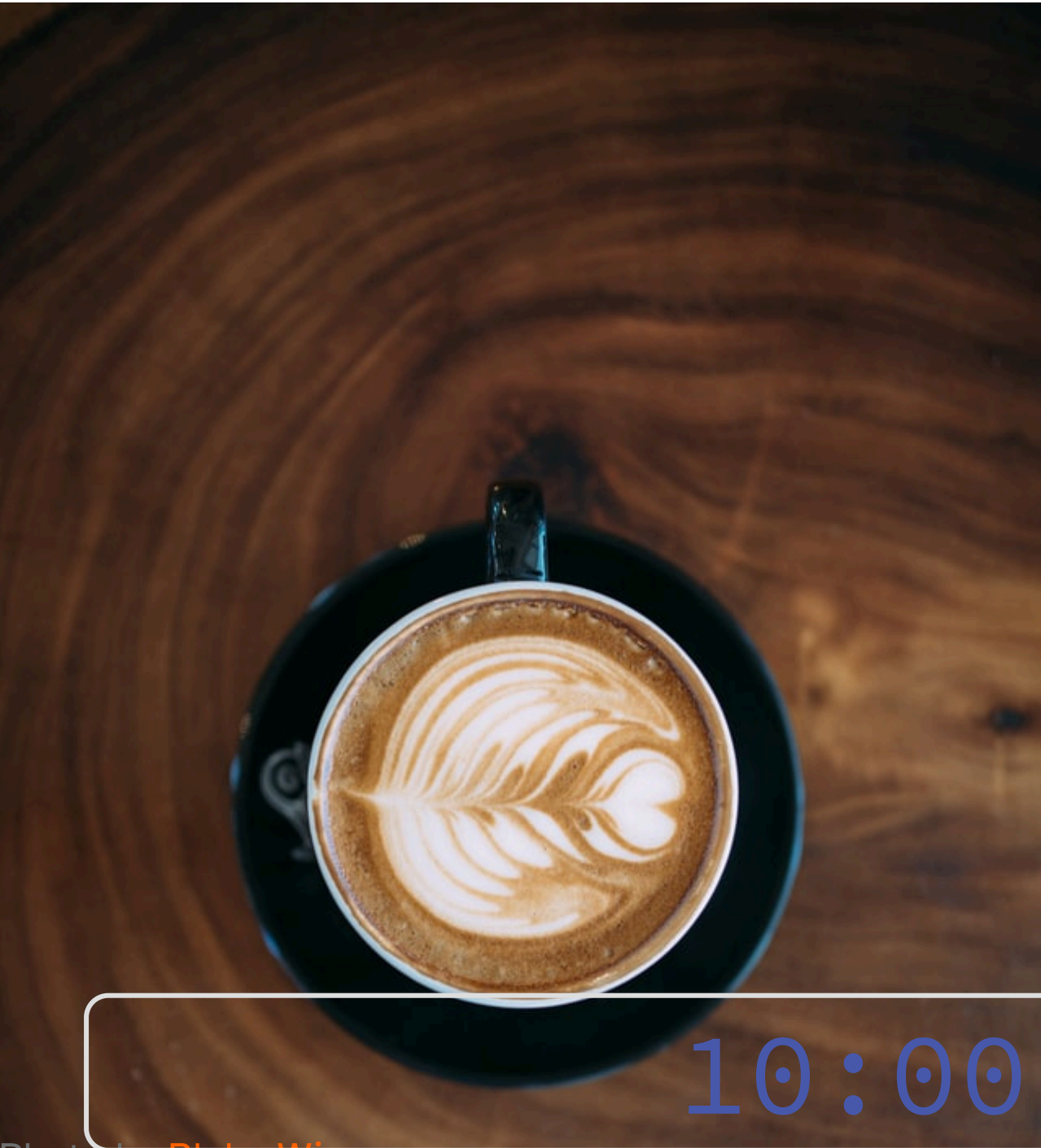


# live-data-transformation

 Follow along on the screen

1. Open the GitHub organisation for the course: <https://github.com/cven5999-ss25>
2. You will find a repository titled: **wk-04-USERNAME** (with your GitHub Username)
3. You will “clone” this repository to Posit Cloud

# Break



10:00

Photo by [Blake Wisz](#)

[cven5999-ss25.github.io/website/](https://cven5999-ss25.github.io/website/)

# Pair Programming Exercise

# Pair Programming Exercises

- Two learners work together in a break out session
- One person (the driver) shares the screen and does the typing
- The other person (the navigator) offers comments and suggestions
- Roles get switched

# hw-data-transformation

1. Head over to posit.cloud
2. Open the workspace for the course [cven5999-ss25](#)
3. Open “Content”
4. Open your project
5. Open the [hw-data-transformation.qmd](#) file
6. Start working through the exercises in the file together

# Homework week 4

# Homework due dates

- All material on [course website](#)
- Homework assignment & learning reflection due: **2025-06-27**



# Thanks! 🌻

Slides created via revealjs and Quarto:

<https://quarto.org/docs/presentations/revealjs/>

Access slides as [PDF on GitHub](#)

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